

Advice Moments

A governed personalization layer that surfaces the right advice step, to the right self-directed investor, at the right moment.

Scope & discretion. Independent thinking exercise built entirely from public information. Not affiliated with, endorsed by, or based on confidential information from Vanguard. Product names are public; all figures, thresholds, and timelines are illustrative and directional, intended to frame a conversation.

1 · PROBLEM

For a self-directed investor, advice is still a destination, a separate decision and motion, when it should be a moment. The investor who would benefit most, navigating a rollover, a windfall, a new goal, or simply rising complexity, rarely gets a relevant, plain-language prompt while the need is live. The capability to detect that moment and respond to it well, in-product, at scale, and within fiduciary guardrails, is the opportunity.

Vanguard's investor-owned structure makes broadening access to advice both mission-aligned and a growth lever, and leadership has publicly framed it as a priority. The largest pool of movable assets and unmet need sits at the first transition of the advice continuum: self-directed investors never matched to the next appropriate step.

2 · GOALS & NON-GOALS

GOALS

- Grow appropriately advised assets via well-timed, relevant prompts
- Match each investor to the genuinely suitable step (including staying self-directed)
- Ship value fast with no model risk, then add intelligence behind evals
- Make every prompt suitable, explainable, capped, and reversible

NON-GOALS

- Maximizing clicks or up-tier conversions for their own sake
- Specific security recommendations or performance promises
- Replacing advisors; this routes to them with context
- A persistent "get advice" ad rather than a moment-triggered prompt

3 · USERS & JOBS-TO-BE-DONE

USER	JOB-TO-BE-DONE	WHAT "GOOD" LOOKS LIKE
Self-directed investor	Handle a change in my situation without becoming an expert	A relevant, honest, optional prompt at the right time
Advice-curious investor	Understand whether advice fits, and at what cost	Plain-language fit + fee, one tap to a consult or digital start
Advisor / advice ops	Receive qualified, in-context consults I can absorb	Clean handoff with the triggering context; demand it can serve
Compliance / risk	Ensure every prompt is suitable and defensible	Pre-send suitability gate, audit trail, documented governance

4 · SOLUTION OVERVIEW

A closed loop: **Signal** (detect a real moment) → **Score** (would advice benefit, which tier) → **Nudge** (governed, explainable prompt) → **Route** (human consult or digital flow). Outcomes feed evals that re-tune scoring, copy, and frequency. The system should become both more useful and more careful over time.

5 · FUNCTIONAL REQUIREMENTS (PHASED)

Phase 0 · Rules & library (~1 quarter, no ML)

- Deterministic triggers from public, first-party signals (e.g., rollover detected, cash above a threshold, a goal added)
- Curated library of plain-language prompts, each pre-mapped to a suitable advice tier and pre-cleared by compliance
- One surface, one or two moments, strict frequency caps and cool-downs
- Explainability ("why am I seeing this") and one-tap opt-out from day one

Phase 1 · Predictive scoring (+1–2 quarters)

- Propensity-and-benefit model ranks who, which tier, and when
- Holdout groups measure true incremental lift, not clicks
- Suppression model prevents over-contacting; release gated by evals

Phase 2 · Generative guidance (later)

- LLM explains the recommendation and answers follow-ups in plain language
- Strictly retrieval-grounded; no performance promises; no specific security advice
- Human-in-the-loop for anything approaching a recommendation boundary

6 · THE BEHAVIORAL CONTRACT

The recommendation is probabilistic: the same investor can be scored differently as data shifts, and quality lives in distributions, not pass/fail. So the spec defines what good looks like numerically, how the system fails safely, and where it must stop.

6.1 Eval-driven acceptance criteria (illustrative)

DIMENSION	HOW MEASURED	RELEASE BAR
Suitability precision	Audited sample + rules check on every prompt	≥ 99.5%; zero tolerance for clearly unsuitable
Incremental benefit	Holdout vs. nudged; appropriate take-up lift	Positive, significant over control
Helpfulness	Human-rated relevance and clarity	≥ 4.3 / 5 median
Non-coercion / tone	LLM-as-judge + human review for pressure/hype	0 coercive/misleading in sample
Factuality (Phase 2)	Grounded-generation eval; unsupported-claim rate	< 0.5%, block on fail

Ship in reporting mode first; compare automated scores to human judgment; then block deployment on trusted metrics for safety, suitability, and regression cases.

6.2 Failure modes

FAILURE	DETECT	USER-FACING RESPONSE	ALERT & RECOVER
Over-nudging	Caps; dismissal/complaint rate by cohort	Suppress; cool-down; "show me less"	Spike alerts; auto-throttle; review copy/targeting
Unsuitable recommendation	Pre-send suitability gate; audit sample	Hard block; nothing unsuitable ships	Gate bypass pages team; roll back; re-eval
Model drift	Live score-distribution & lift vs. baseline	Fall back to Phase 0 rules	Drift breach on-call; retrain; staged re-release
Generative hallucination	Grounded evals; unsupported-claim classifier	Constrain to retrieved facts; refuse off-scope	Rate over bar blocks deploy; human review queue

6.3 Guardrails & governance

Reg BI / fiduciary suitability gate

Human-in-the-loop at recommendation boundaries

Frequency caps & cool-downs

Explainability ("why am I seeing this")

One-tap opt-out, honored everywhere

No performance promises / no security advice

EU AI Act-style risk classification & documentation

Bias & fairness review across cohorts

6.4 Data & model dependencies

- First-party, consented signals only; clear lawful basis and retention limits
- Suitability reference data and rules owned with compliance
- Model registry, versioning, and rollback; eval suite in CI before any release
- Phase 2: retrieval corpus restricted to approved, current, factual content

7 · SUCCESS METRICS

- **North star:** appropriately advised assets (incremental vs. holdout)
- **Funnel:** relevant nudge → consult booked / digital start → enrolled
- **Guardrails (must not regress):** suitability precision, complaint rate, opt-out rate
- **Business:** downstream retention and advised vs. self-directed mix

8 · GO-TO-MARKET & ROLLOUT

- Start on the highest-intent logged-in surface, one or two moments, small holdout-controlled cohort
- Coordinate with advice operations so consult demand matches capacity
- Expand surfaces and moments only as evals and guardrail metrics hold

9 · RISKS & MITIGATIONS

RISK	MITIGATION
Trust erosion from feeling "sold to"	Moment-triggered relevance, honesty about fee/fit, hard caps, easy opt-out
Compliance throughput bottleneck	Pre-cleared library in Phase 0; templated, auditable prompts
Advisor capacity can't absorb demand	Throttle by capacity; favor digital routes where suitable
Optimizing the wrong metric	North star is incremental suitable assets, not clicks; guardrails gate release

10 · OPEN QUESTIONS FOR THE TEAM

- Where does the activation gap leak most: public site, logged-in, onboarding, or mobile?
- Real constraint on advice growth: surfacing, advisor capacity, or compliance throughput?

- How is suitability operationalized today, and what must a pre-send gate integrate with?
- What take-up number does the team already trust, to anchor evals?
- How will the India product org own this: end-to-end surfaces, or with US client-experience owners?

Sources & notes. Built from public information on Vanguard's business, advice continuum, and investor-owned structure; public statements on broadening access to advice; U.S. SEC Regulation Best Interest; the EU AI Act's risk-based approach to AI governance; and current public practice on AI product evals and failure-mode design. All figures, thresholds, and timelines are illustrative. Independent work by Nayan Lal, not affiliated with or endorsed by Vanguard, and not based on any confidential information. nayanlal1909@gmail.com · [linkedin.com/in/nayan-lal](https://www.linkedin.com/in/nayan-lal)